

GIORGIO MORRA

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*US Work Status: F-1 Student **Eligible for OPT***

INTRODUCTION

Mechanical Engineering graduate student from Politecnico di Milano specializing in Mechatronics and Robotics, currently conducting Master's Thesis research at NASA JPL. I aim to contribute to innovative solutions for the control and navigation of Autonomous Systems.

EDUCATION

M.Sc. in Mechanical Engineering (Mechatronics and Robotics) 2023 – Ongoing

Politecnico di Milano, Italy

International Mobility Extra-EU: Pontificia Universidad Javeriana, Colombia

B.Sc. in Mechanical Engineering 2020 – 2023

Politecnico di Milano, Italy

Erasmus+ Program: Universidad Pontificia Comillas, Spain

EXPERIENCE

Robotics Navigation & State Estimation Intern *Dec 2025 – Ongoing*

NASA Jet Propulsion Laboratory – Robotic Mobility and Manipulation Section Pasadena, CA

Contributed to the **Venus Variable Altitude Aerobot** project, focusing on localization strategies for the Venusian atmosphere.

- Modeled the balloon's flight dynamic and an interplanetary (Venus-Earth-Sun) communication model;
- Simulated onboard measurements for sun sensors, radar altimeters, and accelerometers;
- Designed the state prediction model and selected optimal filtering techniques for trajectory reconstruction;
- Performed sensitivity analyses to assess state estimation accuracy under varying sensor availability and noise levels;
- Developed a high-fidelity thermal FEM model to simulate the thermodynamics of the external balloon.

UNIVERSITY PROJECTS

Path Planning, Optimal Control and Estimation

- Solved non-linear optimal control problems for shuttle dynamics, successfully applying both indirect and direct numerical methods;
- Designed and tuned a Linear Quadratic Regulator (LQR) to accurately control an electromechanical actuator and implemented an Extended Kalman Filter (EKF) for robust state and parameter estimation;
- Developed an A* path-finding algorithm on Python for autonomous UAV navigation in complex urban environments, ensuring strict obstacle avoidance and low computational cost;
- Developed a high-level Model Predictive Control on Python for dynamic point tracking, incorporating real-time constraints for moving-obstacle avoidance.

Collaborative Elbow Exoskeleton for Stroke Rehabilitation

- Developed a 1-DoF collaborative elbow exoskeleton to support flexion-extension rehabilitation for post-stroke patients;

- Implemented a "teach-and-play" control strategy, a force sensor, and a servo motor to automate therapeutic motions;
- Designed admittance control for transparent trajectory recording and passive control for accurate, autonomous movement playback;
- Modeled system kinematics and dynamics, successfully validating requirements for motion smoothness, tracking precision, and user safety.

SCARA Robotic Arm Simulation & Motion Planning

- Developed a comprehensive kinematic and dynamic simulation model for a SCARA industrial robotic arm;
- Mapped the operational workspace and analyzed manipulability ellipses to evaluate the robot's kinematic dexterity and performance;
- Designed and optimized motion planning algorithms, comparing the computational efficiency and smoothness of trajectories generated via lines, parabolas, and splines;
- Formulated a multi-body dynamic model to accurately estimate joint torques under varying payload conditions.

SKILLS

Prog. Languages: Python, Matlab, C++

Simulation Tools: Simulink, Simscape

CAD Software: Inventor, SolidWorks, Solid Edge, CATIA V5

Various: L^AT_EX, Microsoft Office

LANGUAGES

Italian (Native) — English (C1) — Spanish (C1)